SIMATS ENGINEERING
ASSESSMENT TOOL1

COURSE CODE: MLA 02 **COURSE NAME: Fundamentals Of Machine Learning**

COURSE OUTCOME COVERED

CO2: *Apply supervised learning algorithms for regression, classification, and predictive modeling tasks to solve structured data problems in practical applications.* *(BL3)*

Assessment Tool Weightage: 40%

QUESTIONS: 10 **Total Marks:50**

Annexure A

Analytical Problem Solving

Code of Conduct:

I ***R.Pranathi(192525076)*** *certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: R. Pranathi

Annexure A

Analytical Problem Solving

Q1.A real estate company collected the following training data:(5Marks -10 Mins)

House No	Area (sq.ft)	Bedrooms	Price (₹ Lakhs)
1	1000	2	50
2	1500	3	75
3	800	2	40
4	1200	3	60
5	2000	4	90

(i) Construct a linear regression model of the form

$$Price = \beta_0 + \beta_1(Area) + \beta_2(Bedrooms)$$

(ii) Estimate the regression coefficients.

(iii) Interpret the meaning of each coefficient.

Q2.An email system uses the following feature values:(5Marks -10 Mins)

Email	Word Frequency (Offer)	Word Frequency (Win)	Spam (1/0)
1	3	2	1
2	2	1	1
3	1	0	0
4	3	3	1
5	0	1	0

(i) Construct a linear classification model.

(ii) Determine the decision boundary.

(iii) Classify a new email with: Offer = 2, Win = 1.

Q3.A medical dataset is given:(5Marks -10 Mins)

Patient	Fever (1/0)	Headache (1/0)	Disease (Yes/No)
1	1	1	Yes
2	1	0	Yes

Patient	Fever (1/0)	Headache (1/0)	Disease (Yes/No)
3	0	1	No
4	1	1	Yes
5	0	0	No

(i) Compute prior probabilities.

(ii) Compute conditional probabilities.

(iii) Estimate the probability that a patient with Fever = 1 and Headache = 0 has the disease.

Q4.Given the following dataset:(5Marks -10 Mins)

Data Point	X_1	X_2	Class
1	2	3	C1
2	3	4	C1
3	5	6	C2
4	6	7	C2
5	4	5	C2

(i) Construct a discriminant function:

$$g(x) = w_1 x_1 + w_2 x_2 + w_0$$

(ii) Determine the decision boundary separating the two classes.

(iii) Classify a new point ($X_1 = 3$, $X_2 = 3$).

Q5.A telecom company collected the following data:(5Marks -10 Mins)

Customer	Monthly Charges	Tenure (Months)	Churn (1/0)
1	500	12	0
2	700	8	1
3	400	15	0
4	900	5	1
5	600	10	1

(i) Formulate the Bayesian logistic regression model.

(ii) Estimate model parameters using the dataset.

(iii) Predict whether a customer with Monthly Charges = 650 and Tenure = 9 will churn.

Q6.Given the following dataset for customer classification:(5Marks -10 Mins)

Customer	Age	Income	Class
C1	25	40	A
C2	30	60	A
C3	35	80	B
C4	40	20	B

For a new customer (Age = 32, Income = 50):

(i).Compute Euclidean distance from all points

(ii).Classify using k = 3

(iii).Determine the final class label

Q7.Given the dataset:(5Marks -10 Mins)

Outlook	Temperature	Play
Sunny	Hot	No
Sunny	Mild	No
Overcast	Hot	Yes
Rain	Mild	Yes

(i) . Compute Entropy of Play

(ii) . Calculate Information Gain for attribute Outlook

(iii). Identify the root node of the decision tree

Q8.Given data points:(5Marks -10 Mins)

X1	X2	Class
2	2	+1
4	4	+1
2	4	-1
4	2	-1

(i) .Plot the points

(ii) .Determine a linear separating hyperplane

(iii).Find the equation of the decision boundary

Q9.Given confusion matrices:(5Marks -10 Mins)

Logistic Regression:

	Pred +	**Pred -**
Actual +	40	10
Actual -	5	45

Naïve Bayes:

	Pred +	**Pred -**
Actual +	35	15
Actual -	10	40

(i).Compute Accuracy, Precision, Recall for both models

(ii).Compare results and identify the better model

Q10.Given points:(5Marks -10 Mins)

(2, 4), (3, 5), (8, 9), (9, 10)

Initial centroids:

C1 = (2,4), C2 = (8,9)

(i).Perform one iteration of K-means

(ii).Assign clusters

(iii).Compute new centroids

RUBRICS FOR CALCULATION

Numerical Problems

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

1ANS:

A real estate company collected the following training data.

House No.	Area (sq. ft.)	Bedrooms	Price (₹ Lakhs)
1	1000	2	50
2	1500	3	75
3	800	2	40
4	1200	3	60
5	2000	4	90

(i) Construct a linear regression model

A multiple linear regression model can be written as:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2$$

Where:

- Y = Predicted house price in lakhs.
- X_1 = Area of the house in square feet.
- X_2 = Number of bedrooms.
- β_0 = Intercept.
- β_1 = Regression coefficient for area.
- β_2 = Regression coefficient for bedrooms.

(ii) Estimate the regression coefficients

The coefficients are estimated by minimizing the sum of squared errors:

$$SSE = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

For the given dataset, the least-squares estimates are approximately:

$$\beta_0 = 3.524 \; ; \; \beta_1 = 0.0461 \; ; \; \beta_2 = -2.744$$

Therefore, the fitted regression equation is:

$$\hat{y} = 3 \cdot 524 + 0.0461 X_1 - 2.744 X_2$$

(iii) Interpret each coefficient

1. Intercept (β_0= 3.524)

The intercept represents the predicted house price when the area and number of bedrooms are both zero. This is mathematically useful but has little practical meaning for a real house.

2. Area coefficient (β_1= 0.0461)

Keeping the number of bedrooms constant, an increase of one square foot is associated with an estimated increase of approximately ₹0.0461 lakh in house price.

For an increase of 100 square feet:

$$100(0.0461) = 4.61$$

The estimated price increases by ₹4.61 lakhs, holding the bedroom count constant.

3. Bedroom coefficient ($\beta_2 = -2.744$)

Keeping the area constant, one additional bedroom is associated with an estimated decrease of ₹2.744 lakhs.

This negative coefficient is a consequence of the small dataset and the relationship between area and bedroom count. It should not be interpreted as proof that adding a bedroom reduces the actual market price.

Result: The multiple linear regression model has been constructed, and its coefficients have been estimated using the least-squares method.

2ANS:

The dataset is given below.

Email	Offer Frequency (X1X_1)	Win Frequency (X2X_2)	Spam (YY)
1	3	2	1
2	2	1	1
3	1	0	0
4	3	3	1
5	0	1	0

(i) Construct a linear classification model

A linear classification model can be represented as:

$$z = b_0 + b_1 x_1 + b_2 x_2$$

For a simple linear classifier, we can choose:

$$z = x_1 + x_2 - 2$$

The classification rule is:

$$\hat{y} = \begin{cases} 1, z > 0 \\ 0, z \leq 0 \end{cases}$$

Here, Y=1 represents spam, and Y=0 represents non-spam.

This is a possible linear classifier for the given training data.

(ii) Determine the decision boundary

The decision boundary occurs when:

$$Z = 0$$

Substituting the model:

$$x_1 + x_2 - 2 = 0$$

$$x_1 + x_2 = 2$$

Or:

$$x_2 = 2 - x_1$$

This straight line separates the spam and non-spam classes for the selected classifier.

(iii) Classify a new email with Offer = 2 and Win = 1

Given:

$$x_1 = 2, \qquad x_2 = 1$$

Substitute into the model:

$$Z = 2 + 1 - 2$$

$$Z = 1$$

Since z > 0,

$$\hat{y} = 1$$

Result: The new email is classified as **spam**.

3ANS:

The medical dataset is given below.

Patient	Fever	Headache	Disease
1	1	1	Yes
2	1	0	Yes
3	0	1	No
4	1	1	Yes
5	0	0	No

(i) Compute prior probabilities

There are five patients.

Three patients have the disease, and two do not.

The prior probability of disease is:

$$P(\text{Disease} = \text{Yes}) = \frac{3}{5}$$

$$P(\text{Yes}) = 0.6$$

The prior probability of no disease is:

$$P(\text{Disease} = \text{No}) = \frac{2}{5}$$

$$P(\text{No}) = 0.4$$

(ii) Compute conditional probabilities

For patients with disease

Among the three patients with disease, all three have fever.

$$P(\text{Fever} = 1 \mid \text{Yes}) = \frac{3}{3} = 1$$

Two of these three patients have a headache.

$$P(\text{Headache} = 1 \mid \text{Yes}) = \frac{2}{3}$$

One patient has no headache.

$$P(\text{Headache} = 0 \mid \text{Yes}) = \frac{1}{3}$$

For patients without disease

Among the two patients without disease, neither has fever.

$$P(\text{Fever} = 1 \mid \text{No}) = \frac{0}{2} = 0$$

Both patients without disease have no fever.

$$P(\text{Fever} = 0 \mid \text{No}) = 1$$

One patient has a headache.

$$P(\text{Headache} = 1 \mid \text{No}) = \frac{1}{2}$$

One patient has no headache.

$$P(\text{Headache} = 0 \mid \text{No}) = \frac{1}{2}$$

The conditional probabilities can be summarized as follows:

Probability	Disease = Yes	Disease = No
P(Fever = 1 | Disease)	1	0
P(Fever = 0 | Disease)	0	1
P(Headache = 1 | Disease)	$\frac{2}{3}$	$\frac{1}{2}$
P(Headache = 0 | Disease)	$\frac{1}{3}$	$\frac{1}{2}$

(iii) Estimate the probability for Fever = 1 and Headache = 0

Given:

$$\text{Fever} = 1, \qquad \text{Headache} = 0$$

Using the Naïve Bayes assumption:

$$P(\text{Yes} \mid F,H) \propto P(\text{Yes})\, P(F = 1 \mid \text{Yes})\, P(H = 0 \mid \text{Yes})$$

Substituting:

$$P(\text{Yes} \mid F,H) \propto 0 \cdot 6 \times 1 \times \tfrac{1}{3} = 0.2$$

For the No class:

$$P(\text{No} \mid F,H) \propto P(\text{No})\, P(F = 1 \mid \text{No})\, P(H = 0 \mid \text{No})$$

$$= 0 \cdot 4 \times 0 \times \tfrac{1}{2}$$

$$= 0$$

Normalizing the probabilities:

$$P(\text{Yes} \mid F,H) = \frac{0 \cdot 2}{0 \cdot 2 + 0}$$

$$P(\text{Yes} \mid F = 1, H = 0) = 1$$

Therefore:

$$P(\text{Disease} = \text{Yes} \mid F = 1, H = 0) = 100\%$$

Result: According to the given training dataset and the Naïve Bayes model, the patient is classified as **Disease = Yes**.

4ANS:

The dataset is given below.

Data Point	X1X_1	X2X_2	Class
1	2	3	C1
2	3	4	C1
3	5	6	C2
4	6	7	C2
5	4	5	C2

(i) Construct a discriminant function

The discriminant function is written as:

$$g(X) = \omega_1 x_1 + w_2 x_2 + b$$

For the given data, observe that:

- Both C1 points satisfy $x_2 = x_1 + 1$
- All C2 points also satisfy $x_2 = x_1 + 1$

Therefore, **the two classes lie on exactly the same line**.

This means that the dataset does not provide a unique linear boundary that separates C1 and C2. A standard linear discriminant analysis model with a shared covariance matrix also encounters a singular covariance matrix because the two features are perfectly correlated.

A unique, statistically justified discriminant function cannot be estimated from this dataset as written.

For an illustrative classification rule, one could choose a threshold on X1X_1:

$$g(X) = x_1 - 3.5$$

Then classify:

$$\hat{C} = \begin{cases} C_1, & g(x) < 0 \\ C_2, & g(x) > 0 \end{cases}$$

This threshold correctly classifies the two C1 observations and all three C2 observations.

(ii) Determine the decision boundary

Using the illustrative function:

$$x_1 - 3.5 = 0$$

$$x_1 = 3.5$$

This is a vertical line separating the two classes in the given sample.

(iii) Classify the new point (3,3)

Given:

$$x_1 = 3, \qquad x_2 = 3$$

Calculate:

$$g(X) = 3 - 3.5 = -0.5$$

Since $g(X) < 0$

$$\hat{c} = c_1$$

Result: Under the illustrative threshold rule, the new point is classified as **C1**. However, the supplied data do not uniquely determine a linear discriminant function because the class points overlap on the same line.

5ANS:

The dataset is given below.

Customer	Monthly Charges	Tenure (Months)	Churn
1	500	12	0
2	700	8	1
3	400	15	0
4	900	5	1
5	600	10	1

(i) Formulate the Bayesian logistic regression model

Logistic regression estimates the probability that a customer will churn.

Let:

- x_1= Monthly charges.
- x_2 = Tenure.
- Y = Churn (1 = Yes, 0 = No).

The logistic regression model is:

$$P(Y = 1 \mid X) = \frac{1}{1 + e^{-z}}$$

where:

$$z = \beta_0 + \beta_1 x_1 + \beta_2 x_2$$

Therefore:

$$P(\text{Churn} = 1 \mid X) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \beta_2 x_2)}}$$

In Bayesian logistic regression, prior distributions are assigned to the model parameters:

$$\beta_j \sim N(0, \sigma^2)$$

The posterior distribution is then proportional to:

$$P(\beta \mid D) \propto P(D \mid \beta)\, P(\beta)$$

Here, $P(D \mid \beta)$ is the likelihood of the observed data, and $P(\beta)$ is the prior distribution.

(ii) Estimate model parameters using the dataset

The five observations provide the training data for estimating the coefficients.

The log-likelihood is:

$$\ell(\beta) = \sum_{i=1}^{n} [\, y_i \log(P_i) + (1 - y_i)\, log(1 - p_i)]$$

where:

$$P_i = \frac{1}{1+e^{-(\beta_0+\beta_1 x_{1i}+\beta_2 x_{2i})}}$$

In Bayesian estimation, the likelihood is combined with the prior to obtain the posterior distribution.

The exact coefficient values cannot be determined uniquely from the question alone, because the prior distributions, prior variances, feature scaling, and parameter-estimation procedure have not been specified.

Therefore, numerical coefficients should not be invented.

(iii) Predict churn for Monthly Charges = 650 and Tenure = 9

Given:

$$x_1 = 650, \;\; x_2 = 9$$

The predicted probability is:

$$P(\text{Churn} = 1) = \frac{1}{1+e^{-(650\beta_1+9\beta_2+\beta_0)}}$$

The decision rule is:

$$\hat{y} = \begin{cases} 1, & P(churn = 1) \geq 0 \cdot 5 \\ 0 & , p(churn = 1) < 0 \cdot 5 \end{cases}$$

Result: The Bayesian logistic regression model has been formulated. Numerical estimation and the final churn prediction require a specified prior and an estimation procedure.

6ANS:

The dataset is given below.

Customer	Age	Income	Class
C1	25	40	A
C2	30	60	A
C3	35	80	B
C4	40	20	B

New customer:

$$X = (32, 50)$$

We need to calculate the Euclidean distances and classify the new customer using k=3k=3.

(i) Compute Euclidean distance from all points

The Euclidean distance formula is:

$$d = \sqrt{(x_1 - x_{1i})^2 + (x_2 - x_{2i})^2}$$

Distance from C1 = (25, 40)

$$d_1 = \sqrt{(32 - 25)^2 + (50 - 40)^2}$$

$$= \sqrt{49 + 100}$$

$$= \sqrt{149}$$

$$d_1 = 12 \cdot 21$$

Distance from C2 = (30, 60)

$$d_2 = \sqrt{(32 - 30)^2 + (50 - 60)^2}$$

$$= \sqrt{4 + 100}$$

$$= \sqrt{104}$$

$$d_2 = 10 \cdot 20$$

Distance from C3 = (35, 80)

$$d_3 = \sqrt{(32 - 35)^2 + (50 - 80)^2}$$

$$= \sqrt{9 + 900}$$

$$= \sqrt{909}$$

$$d_3 = 30 \cdot 15$$

Distance from C4 = (40, 20)

$$d_4 = \sqrt{(32-40)^2 + (50-20)^2}$$

$$= \sqrt{64 + 900}$$

$$= \sqrt{964}$$

$$d_4 = 31 \cdot 05$$

Customer	Distance	Class
C1	12.21	A
C2	10.20	A
C3	30.15	B
C4	31.05	B

(ii) Classify using k = 3

Arrange the distances in ascending order.

Rank	Customer	Distance	Class
1	C2	10.20	A
2	C1	12.21	A
3	C3	30.15	B

The three nearest neighbours are:

- C2 → Class A
- C1 → Class A
- C3 → Class B

Count the classes:

A = 2, B = 1

(iii) Determine the final class label

Class A has the majority vote.

Final Class = A

Result: Using KNN with k = 3, the new customer with age 32 and income 50 is classified as **Class A**.

7ANS:

The dataset is given below.

Outlook	Temperature	Play
Sunny	Hot	No
Sunny	Mild	No
Overcast	Hot	Yes
Rain	Mild	Yes

(i) Compute the entropy of Play

Entropy measures the impurity or uncertainty in a dataset.

The formula is:

$$H(S) = -\ \sum_i P_i\ log\ 2Pi$$

There are four records:

- Yes = 2
- No = 2

Therefore:

$$P(Yes) = \frac{2}{4} = 0.5$$

$$P(No) = \frac{2}{4} = 0.5$$

Substitute into the entropy formula:

$$H(Play) = -\ [0.5\ log_2(0 \cdot 5) + (0 \cdot 5) log_2(0 \cdot 5)]$$

Since:

$$log_2(0.5) = -1$$

We obtain:

$$H(Play) = -\ [0.5(-1) + 0.5(-1)]$$

$$H(Play) = 1 \text{ bit}$$

(ii) Calculate information gain for Outlook

The formula is:

$$IG(Outlook) = H(Play) - H(Play \mid Outlook)$$

The Outlook attribute has three values.

Sunny

The two Sunny records both have Play = No.

$$H(Sunny) = 0$$

Overcast

The one Overcast record has Play = Yes.

$$H(\text{Overcast}) = 0$$

Rain

The one Rain record has Play = Yes.

$$H(\text{Rain}) = 0$$

The weighted conditional entropy is:

$$H(\text{Play|Outlook}) = \frac{2}{4}(0) + \frac{1}{4}(0) + \frac{1}{4}(0)$$

$$H(\text{Play|Outlook}) = 0$$

Therefore:

$$IG(\text{Outlook}) = 1 - 0$$

$$IG(\text{Outlook}) = 1$$

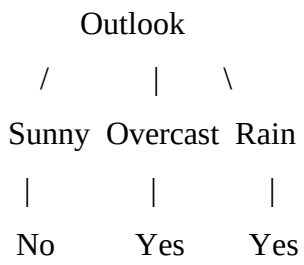
(iii) Identify the root node

The root node is the attribute with the highest information gain.

Since Outlook perfectly separates the classes in this dataset:

Root Node = Outlook

The resulting decision tree is:



Result: The entropy of Play is 1 bit, the information gain of Outlook is 1 bit, and Outlook is selected as the root node.

8ANS:

The given data points are:

x_1	x_2	Class
2	2	+1
4	4	+1
2	4	-1
4	2	-1

(i) Plot the points

The coordinates are:

Positive class (+1):

- (2,2)
- (4,4)

Negative class (-1):

- (2,4)
- (4,2)

A simple coordinate sketch is:

X2

4 -1 +1

(2,4) (4,4)

3

2 +1 -1

(2,2) (4,2)

2 4 X1

The positive points lie on one diagonal, while the negative points lie on the other diagonal.

(ii) Determine a linear separating hyperplane

A linear separating hyperplane in two dimensions has the form:

$$w_1 x_1 + \omega_2 x_2 + b = 0$$

However, the four points form an XOR pattern:

- The two positive points occupy opposite corners.
- The two negative points occupy the other opposite corners.

A single straight line cannot separate the two positive points from the two negative points.

To demonstrate this, suppose a linear function is:

$$f(X) = w_1x_1 + \omega_2x_2 + b$$

For the positive points:

$$f(2,2) > 0$$

$$f(4,4) > 0$$

For the negative points:

$$f(2,4) < 0$$

$$f(4,2) < 0$$

Adding the two positive inequalities gives:

$$6w_1 + 6\omega_2 + 2b > 0$$

Adding the two negative inequalities gives:

$$6w_1 + 6\omega_2 + 2b < 0$$

These conditions contradict each other.

Therefore, a linear separating hyperplane does not exist for this dataset.

(iii) Find the equation of the decision boundary

Because the dataset is not linearly separable, a single linear decision boundary cannot classify all four points correctly.

A nonlinear boundary can be obtained by using:

$$f(X) = (x_1 - 3)(x_2 - 3)$$

The classification rule is:

$$\hat{y} = \begin{cases} +1, & f(x) > 0 \\ -1, & f(x) < 0 \end{cases}$$

This produces the nonlinear decision boundary:

$$(x_1-3)(x_2-3) = 0$$

Equivalently:

$$x_1 = 3 \quad or \quad x_2 = 3$$

This boundary classifies the given points correctly, but it is not a single linear hyperplane. A nonlinear SVM using an appropriate kernel could model this type of separation.

Result: The points are not linearly separable. A linear SVM cannot perfectly classify the given data, but a nonlinear decision boundary can separate the classes.

9ANS:

Logistic Regression

	Predicted Positive	**Predicted Negative**
Actual Positive	40	10
Actual Negative	5	45

Therefore:

- True Positive (TP) = 40
- False Negative (FN) = 10
- False Positive (FP) = 5
- True Negative (TN) = 45

(i) Calculate Accuracy, Precision, and Recall

1. Accuracy

Accuracy measures the proportion of correct predictions.

$$\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN} \times 100$$

$$= \frac{40+45}{40+45+5+10} \times 100$$

$$= \frac{85}{100} \times 100$$

$$\text{Accuracy} = 85\%$$

2. Precision

Precision measures how many predicted positive cases are actually positive.

$$\text{Precision} = \frac{TP}{TP+FP} \times 100$$

$$= \frac{40}{40+5} \times 100$$

$$= \frac{40}{45} \times 100$$

$$\text{Precision} = 88.89\%$$

3. Recall

Recall measures how many actual positive cases are correctly identified.

$$\text{Recall} = \frac{TP}{TP+FN} \times 100$$

$$= \frac{40}{40+10} \times 100$$

$$= \frac{40}{50} \times 100$$

$$\text{Recall} = 80\%$$

Naïve Bayes

	Predicted Positive	**Predicted Negative**
Actual Positive	35	15
Actual Negative	10	40

Therefore:

- TP = 35
- FN = 15
- FP = 10
- TN = 40

1. Accuracy

$$\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN} \times 100$$

$$= \frac{35+40}{35+40+10+15} \times 100$$

$$= \frac{75}{100} \times 100$$

Accuracy = 75%

2. Precision

$$\text{Precision} = \frac{TP}{TP+FP} \times 100$$

$$= \frac{35}{35+10} \times 100$$

$$= \frac{35}{45} \times 100$$

Precision = 77.78%

3. Recall

$$\text{Recall} = \frac{TP}{TP+FN} \times 100$$

$$= \frac{35}{35+15} \times 100$$

$$= \frac{35}{50} \times 100$$

Recall = 70%

Comparison table

Metric	Logistic Regression	Naïve Bayes
Accuracy	85%	75%
Precision	88.89%	77.78%
Recall	80%	70%

(ii) Compare the results and identify the better model

Logistic Regression performs better than Naïve Bayes across all three metrics.

- **Accuracy:** Logistic Regression correctly classifies 85% of the cases, compared with 75% for Naïve Bayes.
- **Precision:** Logistic Regression has higher precision, meaning a greater proportion of its positive predictions are correct.
- **Recall:** Logistic Regression identifies 80% of actual positive cases, compared with 70% for Naïve Bayes.

Therefore:

Better Model: Logistic Regression

Result: Based on the given confusion matrices, Logistic Regression performs better in terms of accuracy, precision, and recall.

10ANS:

Data points:

(2,4), (3,5), (8,9), (9,10)

Initial centroids:

C_1= (2,4)

C_2= (8,9)

(i) Perform one iteration of K-means

K-means involves two main steps:

1. Assign each data point to the nearest centroid.
2. Calculate the new centroid for each cluster by taking the mean of the coordinates of its assigned points.

Step 1: Calculate distances from each point to both centroids

The Euclidean distance formula is:

$$d = \sqrt{(x_1 - Cx)^2 + (x_2 - Cy)^2}$$

Point 1: (2,4)

Distance to c_1= (2,4)

$$d(c_1) = \sqrt{(2-2)^2 + (4-4)^2} = 0$$

Distance to c_2= (8,9)

$$d(C_2) = \sqrt{(2-8)^2 + (4-9)^2}$$
$$= \sqrt{36 + 25}$$
$$= \sqrt{61} = 7 \cdot 81$$

Therefore, (2,4) belongs to Cluster 1.

Point 2: (3,5)

Distance to c_1:

$$d(C_1) = \sqrt{(3-2)^2 + (5-4)^2}$$
$$= \sqrt{1+1}$$
$$= \sqrt{2} = 1 \cdot 41$$

Distance to C_2:

$$d(C_2) = \sqrt{(3-8)^2 + (5-9)^2}$$
$$= \sqrt{25 + 16}$$
$$= \sqrt{41} = 6 \cdot 40$$

Therefore, (3,5) belongs to Cluster 1.

Point 3: (8,9)

Distance to C_1:

$$d(C_1) = \sqrt{(8-2)^2 + (9-4)^2}$$
$$= \sqrt{36 + 25}$$
$$= \sqrt{61} = 7 \cdot 81$$

Distance to c_2:

$$d(C_2) = \sqrt{(8-8)^2 + (9-9)^2} = 0$$

Therefore, (8,9) belongs to Cluster 2.

Point 4: (9,10)

Distance to c_1:

$$d(C_1) = \sqrt{(9-2)^2 + (10-4)^2}$$
$$= \sqrt{49 + 36}$$
$$= \sqrt{85} = 9 \cdot 22$$

Distance c_2:

$$d(C_2) = \sqrt{(9-8)^2 + (10-9)^2}$$
$$= \sqrt{1 + 1}$$
$$= \sqrt{2} = 1 \cdot 41$$

Therefore, (9,10) belongs to Cluster 2.

(ii) Assign clusters

The cluster assignments are:

Data Point	**Distance to C1C_1**	**Distance to C2C_2**	**Assigned Cluster**
(2,4)	0.00	7.81	Cluster 1
(3,5)	1.41	6.40	Cluster 1
(8,9)	7.81	0.00	Cluster 2
(9,10)	9.22	1.41	Cluster 2

Therefore:

Cluster 1={(2,4),(3,5)}

Cluster 2={(8,9),(9,10)}

(iii) Compute new centroids

The new centroid is the mean of the coordinates of the points in each cluster.

New centroid of Cluster 1

Cluster 1 contains:

(2,4), (3,5)

Mean of the first coordinates:

$$C_1 x = \frac{2+3}{2} = 2 \cdot 5$$

Mean of the second coordinates:

$$c_1 y = \frac{4+5}{2} = 4 \cdot 5$$

Therefore:

$$C_1' = (2 \,.\, 5, 4 \,.\, 5)$$

New centroid of Cluster 2

Cluster 2 contains:

(8,9), (9,10)

Mean of the first coordinates:

$$C_{2x} = \frac{8+9}{2} = 8 \cdot 5$$

Mean of the second coordinates:

$$c_{2y} = \frac{9+10}{2} = 9 \cdot 5$$

Therefore:

$$c_2' = (8 \cdot 5, 9 \cdot 5)$$

Result: After one iteration of K-means, the data points are divided into two clusters. The new centroids are (2.5, 4.5) and (8.5, 9.5).